Project Plan

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1. Objective

To design and implement a generic, configurable Demand and Capacity Modelling Platform that enables NHFT to forecast service demand, model capacity, and simulate resource scenarios across its services.

The project will improve data-driven planning, reduce waiting time breaches, and support better workforce utilisation through automated forecasting, capacity modelling, and visual reporting.

2. Stakeholders

Stakeholder	Role / Responsibility	Engagement Level
Yahya Hafeji (Project Lead)	BI Developer responsible for project design, coding, documentation, and delivery.	Responsible
BI Manager (Information Asset Owner)	Oversees data access, security, and governance compliance.	Accountable
Performance Improvement Manager (Project Manager)	Provides project oversight, ensures outputs align with operational goals, and facilitates stakeholder engagement.	Accountable
Service Managers (Pilot Services)	Provide feedback on model outputs and support testing within operational context.	Consulted
Information Governance Team	Ensures compliance with GDPR, DPIA, and data handling policies.	Consulted
University Supervisor	Provides academic guidance and monitors progress towards apprenticeship and module outcomes.	Informed

3. Project Overview and Scope[1]

This project will design, develop, and implement a generic, configurable Demand and Capacity Modelling Platform that can be applied across NHFT's services.

In Scope

- Development of a Python-based modelling framework with modular components for:
    - Data ingestion, validation, and transformation.
    - Statistical time-series forecasting using ETS / Holt-Winters as the dashboard default. (revised)

[1] This revised project scope reflects the final delivered platform, which focused on Python-based data preparation, ETS forecasting, percentile-based demand benchmarking, and a Dash dashboard for interactive analysis (09/03/2026)

    - Service-line demand benchmarking using a configurable percentile framework to generate targets and derived demand ratios. (revised)
    - Capacity reference reporting using staffing snapshots and service-line comparisons, rather than a fully time-aligned workforce model. (revised)
    - A Dash-based visual interface for exploring filters, forecasts, demand metrics, and capacity-related summaries interactively. (revised)
- Creation of configurable dashboard inputs, allowing services and date windows to be analysed without code modification.
- Integration with existing NHFT data sources for patient activity and staffing extracts.
- Power BI version (revised).
- Full-scale deployment across all services beyond the prototype / pilot stage.

Out of Scope

- Machine learning forecasting methods such as Gradient Boosting or XGBoost, which were not part of the delivered forecasting implementation (revised).
- Queueing models and discrete-event simulation which were not implemented in the final project (revised).
- Predictive workforce rostering or AI optimisation beyond the baseline capacity reference view (revised).
- Mobile or external-facing application access.

4. Key Deliverables

The Key Deliverables listed below define the major outputs of the project, providing clarity on what must be produced and when to support delivery.

Deliverable	Description	Expected Output	Deadline
Project Agreement	Formal project approval defining aims, scope, and success metrics.	Signed agreement with NHFT and supervisor.	*15-Oct-25*
Ethics & Legal Assessment	Document detailing GDPR compliance and data protection measures.	Completed DPIA and ethics sign-off.	*05-Nov-25*
Project Approach	Summary of chosen methodology (Kanban), tools, and workflow plan.	Project Approach report.	*19-Nov-25*
Project Plan	Detailed roadmap (this document).	Submission with deliverables and milestones.	*10-Dec-25*
Modelling Platform (MVP)	Prototype system with forecasting and capacity modules.	Functional prototype with documented outputs.	*16-Feb-26*
Dash/ PowerBI Application	Interactive interface to visualise forecasts and capacity scenarios.	Tested version across pilot services.	*28-Feb-26*

Final Presentation	Demonstration and summary of findings to stakeholders.	Presentation and video recording.	*17-Jun-26*

5. Milestones Summary

The Milestones Summary below highlights the project's major deadlines and expected outcomes, ensuring progress can be monitored against defined targets.

Milestone	Target Completion	Success Indicator
Project Documentation & Governance Completed	**Early Dec 2025**	Project Agreement, Project Plan, Methodology, and Ethics & Legal Assessment finalised.
Data Access & Infrastructure Secured	**Mid Dec 2025**	SQL BI warehouse access established; secure environment and governance approvals confirmed.
DPIA & Ethics Approval Confirmed	**Mid Dec 2025**	DPIA and IG sign-off completed; permissions granted for data processing.
Data Engineering & Automation Pipeline (Initial Version)	**Early Jan 2026**	Automated direct-query pipeline created; data validation scripts functioning for pilot services.
Forecasting Module (MVP) Delivered	**Early Feb 2026**	Forecasts generated for three pilot services using validated models.
Capacity & Simulation Module Completed	**Mid Feb 2026**	Simulation engine integrated with capacity modelling; scenario outputs functioning.
Dash/Power BI Application (Beta) Released	**End Feb 2026**	Interactive visual interface accessible, tested, and validated with pilot services.
Final Project Presentation	**Jun-26**	Presentation delivered to stakeholders; solution approved as viable for wider use.

6. Project Methodology & Schedule

Although the project follows a largely linear progression from documentation and governance, through data access, modelling, and application development, it will be managed using Microsoft Planner in a Kanban-style workflow.

The linear structure is reflected in the numbered overarching tasks and deliverables, but Microsoft Planner provides the flexibility to organise, prioritise, and monitor work at the sub-task level, ensuring transparency and continuous progress.

Workstream / Milestone	Dec 2025 (Weeks 1–4)	Jan 2026 (Weeks 1–4)	Feb 2026 (Weeks 1–4)	Deliverable / Outcome
1. Project Setup & Planning				Project Agreement confirmed; Kanban board created;

				stakeholder sign-off.
2. Data Access & Governance				Secure data connections to NHFT SQL warehouse; DPIA approved; governance confirmed.
3. Data Engineering & Automation				Automated data pipelines and validation scripts created; direct-query model established.
4. Forecasting Model Development				Time-series and ML forecasting modules implemented and tested.
5. Capacity & Simulation Modelling				Simulation and queueing components integrated into framework.
6. Dash Application Development				Interactive visualisation interface developed using Dash; initial pilot view deployed.
7. Testing, Validation & Feedback				Forecast validation, stakeholder feedback, and iteration rounds completed.
8. Documentation & Handover				Technical documentation, configuration templates, and OneNote log updates finalised.

The Planner board will include the following Kanban columns:

- **Backlog:** All planned and future tasks, grouped by overarching subjects.
- **Up Next:** Prioritised work items prepared for immediate action.
- **In Progress:** Tasks actively being developed or implemented.
- **In Review:** Work undergoing testing, validation, or stakeholder feedback.

- **Issues:** Any blocked items requiring clarification, rework, or risk mitigation.
- **Completed:** Tasks that meet acceptance criteria and are formally signed off.

Each overarching task (e.g., *Data Access*, *Forecasting Module*, *Simulation Engine*, *Dash App*) will contain subtasks organised using Microsoft Planner checklists.

To support planning and workload management, Fibonacci effort estimates (1, 2, 3, 5, 8, 13…) will be assigned to subtasks. This helps approximate complexity and time investment more realistically than fixed time estimates.

7. RACI Chart

The RACI chart below summarises the overarching project tasks and clarifies who is responsible, accountable, consulted, and informed for each, ensuring clear governance and ownership.

Task / Overarching Workstream	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
1. Project Documentation & Governance	Yahya Hafeji	BI Manager	Supervisor, IG Team	Performance Improvement Manager
2. Data Access & Infrastructure Setup	Yahya Hafeji	BI Manager	IG Team, ICT Support	Performance Improvement Manager
3. Data Engineering & Automation Pipeline	Yahya Hafeji	Project Manager	BI Analysts, Data Engineers	Supervisor
4. Forecasting Module (MVP)	Yahya Hafeji	Project Manager	BI Analysts, Service Leads (Pilot Services)	Supervisor
5. Capacity & Simulation Module	Yahya Hafeji	Project Manager	BI Analysts, Operational Leads	Supervisor
6. Dash / Power BI Application Development	Yahya Hafeji	Project Manager	Service Managers (Pilot Services), UI/UX Advisors	Supervisor
7. Testing, Validation & Refinement	Yahya Hafeji	Project Manager	BI Team, Service Managers	Wider Stakeholders
8. Documentation, Logbook & Final Presentation	Yahya Hafeji	BI Manager	Supervisor	Wider Trust Stakeholders

Notes on Role Definitions

- **Responsible (R):**
  The person doing the work.

- **Accountable (A):**
  Owns the deliverable and signs off the work. (BI Manager for governance/IG-related tasks, Project Manager for technical delivery).
- **Consulted (C):**
  Experts or stakeholders whose input is required (IG, BI Analysts, Service Managers etc.).
- **Informed (I):**
  Stakeholders who need updates but are not directly involved.

8. Risk Assessment

The Risk Assessment below summarises the main project risks and the actions planned to minimise their impact, ensuring proactive management throughout the project.

Risk ID	Description of Risk	Likelihood	Impact	Mitigation / Safeguard
R1	Unauthorised access to pseudonymised data.	Low	High	Access restricted to authorised users. Encrypted network and MFA authentication.
R2	Potential re-identification of individuals from small datasets.	Low	Medium	Aggregation thresholds and suppression rules to limit re-identification of small counts (<5).
R3	Model bias due to incomplete or unbalanced data across services.	Medium	Medium	Regular fairness evaluation and model parameter tuning to maintain equity.
R4	Data extraction or transformation error leading to inaccurate outputs.	Medium	Medium	Automated data validation checks and version-controlled pipelines.
R5	Breach of open-source software licence.	Low	Medium	Use only well-documented libraries under permissive licences (MIT/BSD).
R6	Misinterpretation of model outputs by non-technical users.	Medium	Medium	Include documentation, tooltips, and confidence intervals in dashboards.
R7	Retention of temporary data files outside of secure environment.	Low	High	Implement direct-query automation to remove need for intermediate storage.
R8	Data access delays	Medium	Medium	Obtain early approval from BI Manager and IG team; request test extracts.
R9	Data quality issues	Medium	High	Build automated validation checks and data profiling scripts.
R10	Model bias or poor accuracy	Medium	Medium	Include fairness metrics and parameter tuning to balance performance.
R11	Time constraints due to workload	Medium	High	Use Kanban to manage priorities and allocate fixed weekly project hours.

R12	Technical failure or code loss	Low	Medium	Use Git version control and regular repository backups.
R13	Stakeholder engagement gaps	Medium	Medium	Schedule bi-monthly updates with the Performance Improvement team.

9. Resources and Tools

The Resources and Tools list summarises the platforms and technologies essential for project delivery, supporting effective development and workflow management.

Resource Type	Details
Project Planning	Kanban Board via MS Teams
Programming Environment	Python (Maybe R)
Libraries	Pandas, NumPy, Scikit-learn, Statsmodels, Plotly/Dash
Database Access	NHFT SQL Server - ISEVSQLMIS-REP
Version Control	GitHub
Development Environment	Visual Studio Code (locally) or NHFT-approved virtual desktop
Compute Resources	Local execution for prototypes; potential use of NHFT Virtual Machines or Power BI Service for visualisation
Documentation & Logs	OneNote for logbook and SharePoint for documentation